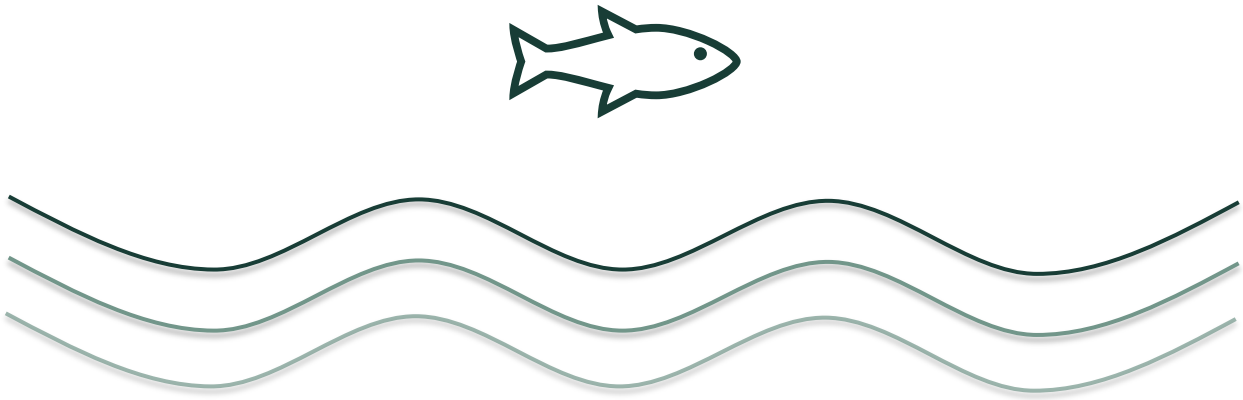
# Fish Population Dynamics in Lake Constance



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**Date**
06 November 2026

**Module**
CO1 – Modeling of complex systems

Abstract ..... 3
Introduction ..... 4
Methods ..... 5
Model Framing ..... 5
System Boundaries ..... 5
Model ..... 6
General Feedback Analysis ..... 7
State Variables ..... 8
Auxiliary Variables ..... 8
ODEs ..... 10
Calibration ..... 10
Data sets ..... 10
Parameters ..... 11
Outputs ..... 11
Model Analysis ..... 12
Numerical stability ..... 12
Degree of validity ..... 13
Sensitivity ..... 13
Results ..... 14
General model behaviour ..... 14
Reproducing the historical data ..... 16
Fishing ban and decline of stickleback population contribution to the whitefish stock recovery ..... 17
Calculation of the maximum annual catch from 2027 onwards to keep the stock above a reference level (50t) until 2050 ..... 18
Discussion ..... 20
Open issues ..... 20
Evidence and implications ..... 20
Learning from the model ..... 21
Literature ..... 22
Figures ..... 24
Tables ..... 24
Appendix ..... 24
Link to GitHub Repository ..... 24
Model as Image ..... 25
Model as Text ..... 26

# Abstract

The whitefish population has long been the most important fish for commercial fishing in Lake Constance. After 2012, however, the population collapsed, and in 2023 only 9.8 tonnes were caught. In 2024, two changes occurred almost simultaneously: a fishing ban came into force, and the population of the invasive stickleback, which competes with whitefish for food and feeds on their eggs and larvae, declined by about 80%. Since then, the whitefish stock has been recovering. However, it remains unclear which change caused this recovery and whether fishing can be resumed sustainably in May 2027. Observational data cannot answer this, because both changes happened at the same time. Therefore, a simplified system dynamics model in Vensim was developed in this project, covering a timespan from 1950 to 2050. It includes zooplankton, young whitefish and adult whitefish, while phosphorus, deep-water oxygen, stickleback pressure and fishing act as external drivers. The model was calibrated against historical catch, survey and zooplankton data. It reproduces the long-term rise of the population until the late 1990s and its subsequent collapse, but not the year-to-year fluctuations. The scenarios show that the fishing ban was the main driver of the short-term recovery, while reduced stickleback pressure is more important in the long run. With a catch of 25 tonnes per year from 2027, the stock only recovers if both changes are in place. If stickleback pressure remains low, up to about 27 tonnes per year can be caught. If it returns, only 5–7 tonnes. These results are scenarios rather than exact forecasts, but they indicate that future catch limits should depend on the development of the stickleback population.

# Introduction

With a surface area of about 536 km², Lake Constance is one of the largest lakes in Central Europe (Bodensee-Openwater, 2023). It is jointly managed by Switzerland, Germany and Austria (Internationale Gewässerschutzkommission für den Bodensee [IGKB], n.d.-b). The whitefish plays a key role in the lake's ecosystem (Ogorelec et al., 2022). At the same time, it has long been the most important food fish and therefore the economic foundation of commercial fishing on Lake Constance (IGKB, 2023; Schweizer Radio und Fernsehen [SRF], 2024).

The development of the whitefish stock is closely linked to the nutrient content of the lake, mainly its phosphorus concentration. From the 1950s onwards, wastewater and agricultural runoff caused phosphorus levels to rise sharply, peaking in the 1970s (IGKB, 2018a). As a result, fishing yields also increased. In 1978, 1200 tonnes of whitefish were caught, the highest annual catch ever recorded (SRF, 2026). Thanks to extensive water protection measures, the phosphorus concentration fell to a natural level of about 6 µg/L over the following decades (IGKB, 2013, 2018a). Consequently, both the whitefish stock and the fishing yields declined again (IGKB, n.d.-a, 2023).

In recent decades, new stressors have emerged. The stickleback, a non-native species, was first recorded in Lake Constance in the 1950s and initially lived in the shallow littoral zone (Wasser-Agenda 21, 2023). In 2012, the stickleback population spread into the open water, where it multiplied rapidly and at times accounted for up to 90% of all fish in the open water (SRF, 2024; Wasser-Agenda 21, 2023). The stickleback competes with the whitefish for zooplankton and feeds on its eggs and larvae (IGKB, 2023; SRF, 2024). Together, these stressors led to a collapse of the whitefish stock (IGKB, 2023). In the year 2023 only 9.8 tonnes were caught (Frei & Kugler, 2026).

In response to the decline of the whitefish population, the International Conference of Authorised Representatives for Lake Constance Fisheries (IBKF) decided to impose a three-year fishing ban on whitefish, starting in January 2024 (Frei & Kugler, 2026; IGKB, n.d.-a). At almost the same time, the stickleback population declined by around 80% for reasons that remain unclear (IGKB, n.d.-a; SRF, 2024). Since then, the whitefish population has shown signs of recovery, with older and larger individuals present in the lake again (Frei & Kugler, 2026; SRF, 2026). From May 2027 onwards, whitefish fishing will be partially permitted again under strict conditions, with a long-term target of 20 to 30 tonnes per year (SRF, 2026).
However, it remains unclear which factor brought about this recovery. Because the fishing ban and the decline of the stickleback coincided, their effects cannot be distinguished based on observational data alone. It also remains to be seen whether a reopened fishery would be sustainable if stickleback pressure returns.

The aim of this project is to develop a simplified system dynamics model of Lake Constance that links phosphorus, plankton, deep-water oxygen and the whitefish population for the period from 1950 to 2050, and to calibrate it using historical measurement data. The model will be used to address the following research questions:

- Is the model able to reproduce the historical data?
- To what extent have the 2024 fishing ban and the decline of the stickleback population each contributed to the recovery of the whitefish stock?
- What is the maximum annual catch that can be permitted from 2027 onwards to ensure that the stock remains above a reference level until 2050, under both low and renewed stickleback pressure?

A simulation model is well suited to answer these questions, as it allows testing scenarios that cannot be observed, such as a fishing ban without the stickleback decline. In addition, system

dynamics captures feedback loops and time delays, such as the maturation time of whitefish or the competition between whitefish and sticklebacks for zooplankton. As the fishing ban expires and fishing is set to partially resume in May 2027 (Frei & Kugler, 2026; SRF, 2026), the results can support decisions on future catch limits.

# Methods

## Model Framing

The model represents a simplified version of Lake Constance that focuses on the whitefish population. It consists of three stocks: zooplankton, young whitefish and adult whitefish.

The whitefish population forms the core of the model. Adult whitefish produce offspring, which are summarised as young whitefish (eggs, larvae and juveniles). After about 2.5 years, the young whitefish mature into adults. Both young and adult whitefish are subject to natural mortality. Part of the adult stock is removed by fishing. Before 2024, a constant share of the adult stock is caught each year. From January 2024 to April 2027, fishing is banned. From May 2027 onwards a fixed annual catch is taken.

How many offsprings survive depends on three factors. The first one is food: the more zooplankton is available, the more young whitefish survive. The second one is deep-water oxygen: whitefish eggs develop on the bottom of the lake over winter and die when the oxygen levels are too low. And last: sticklebacks eat the whitefish eggs and larvae.

Zooplankton is the main food of the whitefish. It grows faster when the phosphorus concentration is higher, because more phosphorus allows more algae to grow. And the zooplankton feeds on these algae. Part of the zooplankton dies naturally, while the rest is eaten by whitefish and sticklebacks. Sticklebacks therefore compete with whitefish for food.

Phosphorus concentration and deep-water oxygen are taken form measurements data. Stickleback pressure is described by an index. It is zero before 2012 and drops by 80% in 2024. From 2027 onwards it either remains low or returns to its former level. In this way different scenarios can be tested to answer the second question.

The model contains three feedback loops. In the reinforcing loop (R1), more adult whitefish produce more offspring, which leads to more adults. In the balancing predator-prey loop (B1), more adult whitefish eat more zooplankton, leaving less food for their offspring. In the balancing loop (B2), a larger adult stock leads to a higher catch value, which reduces the stock.

## System Boundaries

The model represents the whitefish population of Upper Lake Constance. It also includes the zooplankton that the whitefish feed on, the oxygen in the deep water, the sticklebacks and fishing. To keep the model simple, only processes that have a direct effect on the whitefish were considered. Phosphorus and deep-water oxygen are not calculated in the model but taken from measurement data, as they mainly depend on external factors such as wastewater treatment and the water circulation in the winter. Algae were not modelled separately, because their effect is included in the growth of zooplankton. The release of phosphorus from the sediment was neglected, as it only happens when oxygen levels are close to zero. And this has never been observed in lake Constance. The stickleback population itself is not modelled. Only its pressure on the whitefish population has been included as an index, because the reason for its collapse in 2024 is still unknown.
With the time scope of 100 years and an annual time step, seasonal changes within a year have not been modelled. Furthermore, water temperature, stocking, other species such as the quagga mussel and cormorants, as well as the age and size structure of the whitefish were not implemented in the model.

## Model

The model was implemented in Vensim. To make its structure easier to read, the model is presented here in redrawn form (done with Claude). Figure 1 gives an overview of the three stocks, the main external drivers and the feedback loops, while Figures 2, 3 and 4 show the three parts of the model in detail, including all variables and parameters. These figures contain exactly the same variables, connections and equations as the Vensim model but uses a clearer layout and a larger font. The original Vensim model is shown in the Appendix.

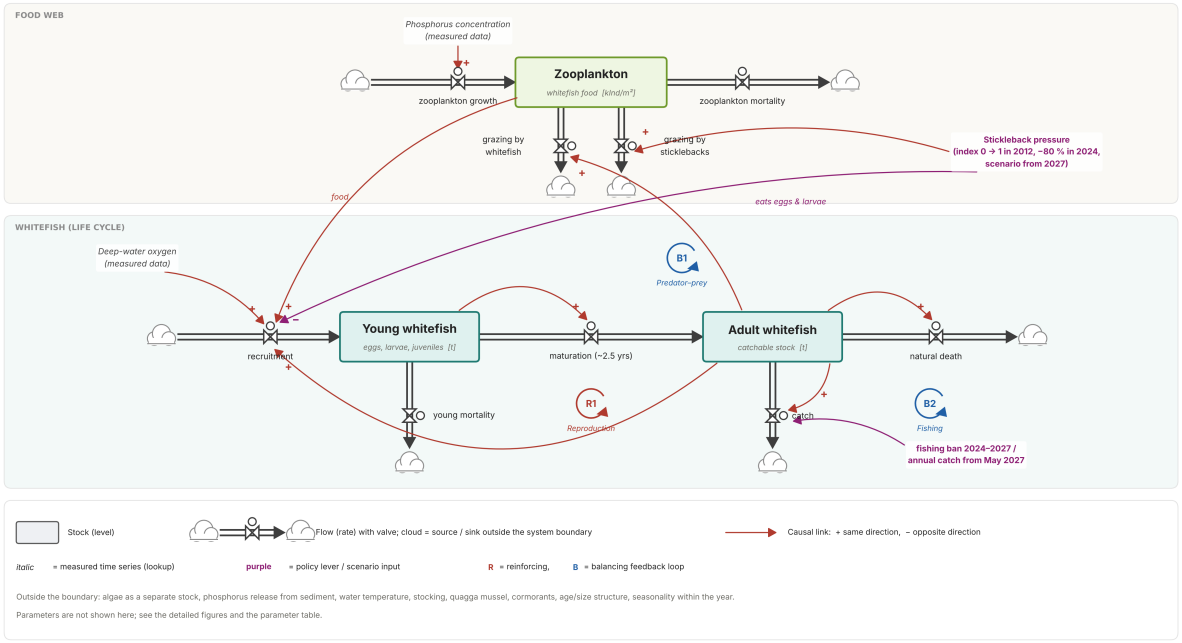



*Figure 1: Model Overview*

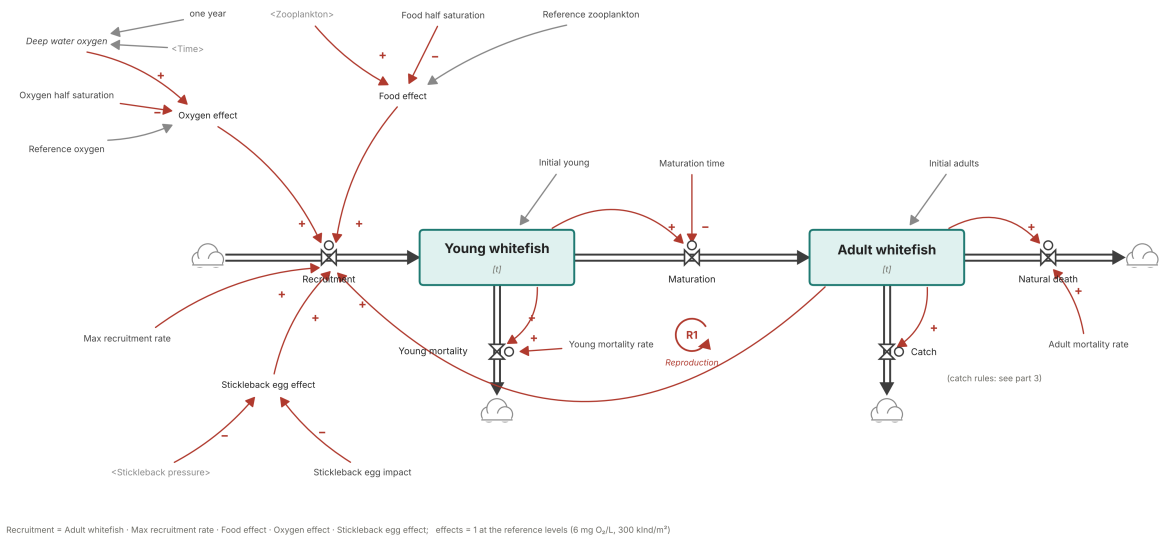



*Figure 2: Whitefish life cycle*

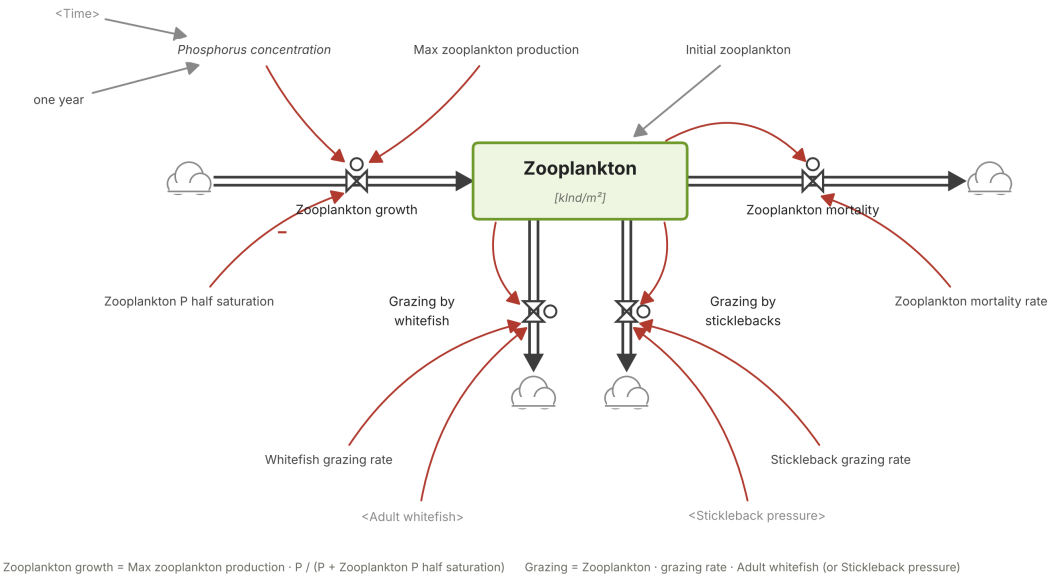



*Figure 3: Zooplankton food web*

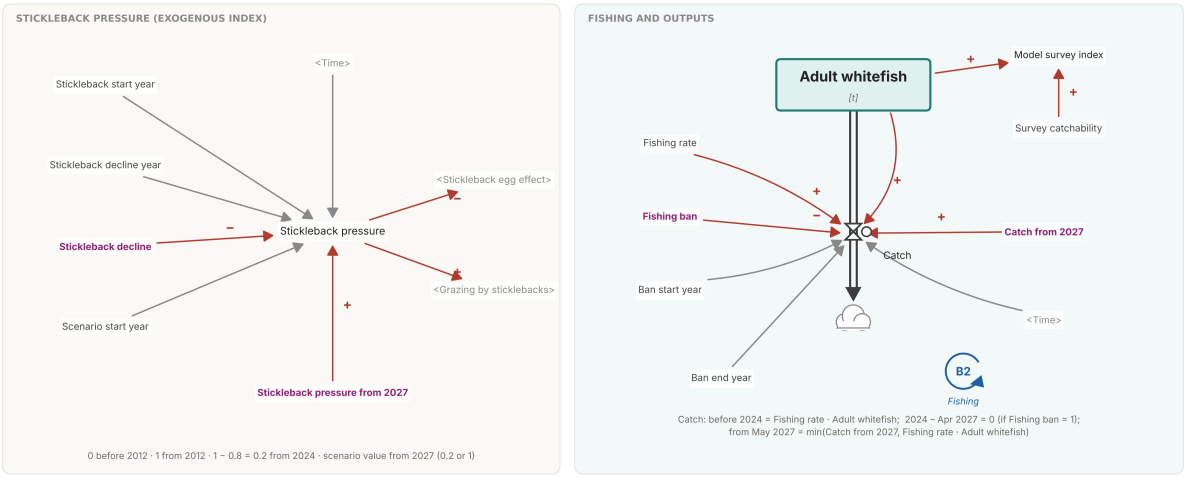



*Figure 4: Stickleback pressure and fishing outputs*

## General Feedback Analysis

The model contains three feedback loops, which are marked in Figures 2, 3 and 4. The first loop is the reinforcing reproduction loop (R1). It connects the adult whitefish with recruitment, the young whitefish and maturation. A larger adult stock produces more offspring and after a delay of about 2.5 years these offsprings mature and increase the adult stock. This loop is responsible for the growth of the population.

The reproduction loop is counteracted by two balancing loops. In the predator-prey loop (B1), a larger whitefish stock consumes more zooplankton. As a result, less food is available for the offspring, which reduces and limits further growth of the population. In the fishing loop (B2), a larger adult stock leads to a higher catch value, which in return reduces the stock.

The relative strength of these loops changes over time. Between 1950 and the 1990s, the reproduction loop dominates as rising phosphorus levels increase the zooplankton and thus the number of offspring. After 2012, the stickleback weakens the reproduction loop by feeding on the whitefish eggs and larvae, while the fishing loop continues to remove adults. This leads to the collapse of the whitefish population in 2024. With the fishing ban in 2024, the fishing

loop is temporarily switched off, and the stickleback pressure decreases, so that the reproduction loop becomes dominant again.

Phosphorus, deep-water oxygen and stickleback pressure act as external factors on these loops. They influence the behaviour of the system but are not affected by it. Therefore, they are not part of any feedback loops themselves.

**State Variables**

| *Variable* | *Input Flows* | *Output Flows* | *Start value* | *Source* |
|---|---|---|---|---|
| Adult Whitefish | Maturation | • Natural death<br>• Catch | 907t | Calibrated against catch (1950-2023) and survey index (1997-2022) |
| Young Whitefish | Recruitment | • Maturation<br>• Young mortality | 1512t | Calculated as equilibrium with the conditions in 1950 |
| Zooplankton | Zooplankton growth | • Zooplankton mortality<br>• Grazing by whitefish<br>• Grazing by stickleback | 332 x 1000 ind./$m^2$ | Calculated as equilibrium with the phosphorus levels in 1950 |

*Table 1: State variables of the model*

The starting value of young whitefish and zooplankton were set so that inflows and outflows are balanced under the conditions of 1950. So that the model starts without artificial jumps.

**Auxiliary Variables**

| *Variable* | *Value / Equation* | *Unit* | *Source* |
|---|---|---|---|
| Food effect | (Zooplankton / (Zooplankton + Food half saturation)) / (Reference zooplankton / (Reference zooplankton + Food half saturation)) | Dmnl | Own formulation; = 1 at reference zooplankton level |
| Oxygen effect | (Deep water oxygen / (Deep water oxygen + Oxygen half saturation)) / (Reference oxygen / (Reference oxygen + Oxygen half saturation)) | Dmnl | Own formulation based on IGKB (2014); = 1 at 6 mg/L |
| Stickleback egg effect | 1 − Stickleback egg impact × Stickleback pressure | Dmnl | Own formulation based on IGKB (2023) and SRF (2024) |
| Stickleback pressure | 0 before 2012; 1 from 2012; 1 − Stickleback decline from 2024; Stickleback pressure from 2027 from 2027 | Dmnl | Wasser-Agenda 21 (2023); SRF (2024); IGKB (n.d.-a) |
| Model survey index | Adult whitefish × Survey catchability | Dmnl | Own formulation, for comparison with survey data |

| *Variable* | *Value / Equation* | *Unit* | *Source* |
|---|---|---|---|
| Phosphorus concentration | Lookup over Time / one year; data 1960–2025; 1950 = 7 (assumption) | µg P/L | IGKB / BOWIS |
| Deep water oxygen | Lookup over Time / one year; data 1963–2025; 1950 = 8 (assumption) | mg $O_2$/L | IGKB |
| Observed catch | Lookup over Time / one year; data 1950–2023 | t/Year | IGKB (read from chart); 2023: Frei & Kugler (2026) |
| Observed survey index | Lookup over Time / one year; data 1997–2022 | Dmnl | IGKB / IBKF |
| Observed zooplankton | Lookup over Time / one year; data 1961–2025 | kInd/m² | IGKB / BOWIS |
| one year | 1 | Year | Technical constant; converts Time into a dimensionless value for the lookup functions |
| Max zooplankton production | 3444.7 | kInd/m²/Year | Calibrated |
| Zooplankton P half saturation | 21.31 | µg P/L | Calibrated |
| Zooplankton mortality rate | 2 | 1/Year | Assumption |
| Whitefish grazing rate | 0.000626 | 1/(t·Year) | Calibrated |
| Stickleback grazing rate | 0.203 | 1/Year | Calibrated |
| Food half saturation | 213.0 | kInd/m² | Calibrated |
| Reference zooplankton | 300 | kInd/m² | Approx. current level (IGKB) |
| Max recruitment rate | 1.051 | 1/Year | Calibrated |
| Oxygen half saturation | 2 | mg $O_2$/L | Assumption |
| Reference oxygen | 6 | mg $O_2$/L | IGKB (2018) |
| Stickleback egg impact | 0.506 | Dmnl | Calibrated |
| Stickleback start year | 2012 | Year | Wasser-Agenda 21 (2023); SRF (2024) |
| Stickleback decline year | 2024 | Year | IGKB (n.d.-a) |
| Stickleback decline | 0.8 | Dmnl | IGKB (n.d.-a); scenario: 0 = no decline |
| Scenario start year | 2027 | Year | Own definition |

| *Variable* | *Value / Equation* | *Unit* | *Source* |
|---|---|---|---|
| Stickleback pressure from 2027 | 0.2 | Dmnl | Scenario: 0.2 = stays low, 1 = returns |
| Maturation time | 2.5 | Year | Assumption |
| Young mortality rate | 0.3 | 1/Year | Assumption |
| Adult mortality rate | 0.25 | 1/Year | Assumption |
| Fishing rate | 0.5 | 1/Year | Assumption |
| Ban start year | 2024 | Year | IBKF; Frei & Kugler (2026) |
| Ban end year | 2027.33 | Year | SRF (2026) |
| Fishing ban | 1 | Dmnl | Scenario: 1 = ban, 0 = no ban |
| Catch from 2027 | 25 | t/Year | SRF (2026), target 20–30 t; scenario |
| Survey catchability | 0.01566 | Dmnl/t | Calibrated |
| Initial adults | 906.7 | t | Calibrated |
| Initial young | 1512.0 | t | Equilibrium with 1950 conditions |
| Initial zooplankton | 331.7 | kInd/m² | Equilibrium with 1950 conditions |

*Table 2: Auxiliary variables of the model*

**ODEs**

| *Variable* | *ODE* |
|---|---|
| Adult Whitefish | $\frac{d\ Adult\ Whitefish}{dt} = Maturation - Natural\ Death - Catch$ |
| Young Whitefish | $\frac{d\ Young\ Whitefish}{dt} = Recruitment - Maturation - Young\ Mortality$ |
| Zooplankton | $\frac{d\ Zooplankton}{dt} = Zooplankton\ Growth - Zooplankton\ Mortality - Grazing\ by\ Whitefish - Grazing\ by\ Sticklebacks$ |

*Table 3: ODE's of the model*

## Calibration

Several parameters of the model could not be taken directly form the literature. These parameters were therefore estimated by calibrating the model against historical measurement data form Lake Constance. The aim of the calibration was to reproduce the long-term development of the whitefish population and its food source between 1950 and 2025.

### Data sets

Three data sets were used to calibrate the model. The annual whitefish catch of the professional fishery in the Upper Lake Constance from 1950-2023 was used as the main reference for the whitefish population (IGKB, n.d.-a). As these values are only published as a chart, they were read from the figure with an estimated accuracy of about ± 10 tonnes. The catch of 2023 was taken form Frei and Kugler (2026). In addition, the whitefish survey index

form 1997-2022 was used as an independent measure of the stock size (IGKB). The density of crustacean zooplankton from 1961-2025 was used to calibrate the food web part of the model (IGKB / BOWIS).

Phosphorus concentration and deep-water oxygen were not used for the calibration but served as model inputs. Since measurements of phosphorus only start in 1960 and those of oxygen in 1963, the values for 1950 were assumed to be 7 µg/L and 8 mg/L respectively. Reflecting the conditions before the onset of eutrophication.

**Parameters**

Nine parameters were calibrated (Table 4). They were estimated by minimising the sum of square differences between the simulated and the observed values of all three datasets. To make the datasets comparable, the differences were scaled to the typical size of each dataset. The survey index and the zooplankton data were given a slightly lower weight than the catch data. The calibration was carried out with a Python implementation of the same model equations, and the resulting parameter values were then transferred to the Vensim model. The survey catchability was not part of the iterative search but was calculated directly as the factor that best scales the simulated adult stock to the observed survey index. The start values of young whitefish and zooplankton were set to equilibrium with the conditions in 1950, so that only the initial adult stock had to be calibrated. All remaining parameters were taken from the literature or set as assumptions.

| *Parameter* | *Lower bound* | *Upper bound* | *Calibrated value* | *Unit* |
|---|---|---|---|---|
| Max recruitment rate | 0.05 | 20 | 1.051 | 1/Year |
| Max zooplankton production | 50 | 100 000 | 3444.7 | 1000 ind./m²/Year |
| Zooplankton P half saturation | 0.5 | 300 | 21.31 | µg P/L |
| Whitefish grazing rate | 0.000001 | 0.05 | 0.000626 | 1/(t·Year) |
| Stickleback grazing rate | 0 | 10 | 0.203 | 1/Year |
| Food half saturation | 10 | 100 000 | 213.0 | 1000 ind./m² |
| Stickleback egg impact | 0 | 1 | 0.506 | Dmnl |
| Initial adults | 50 | 20 000 | 906.7 | t |
| Survey catchability | – | – | 0.01566 | Dmnl/t |

*Table 4: Calibrated parameters used in the model*

## Outputs

| Group | Variable | Unit | Value range (year) |
|---|---|---|---|
| Stocks | Adult whitefish | t | 49 (2024) to 1,436 (1997) |
| | Young whitefish | t | 60 (2024) to 2,705 (1996) |
| | Zooplankton | kInd/m² | 285 (2050) to 1,091 (1977) |
| Flows adult whitefish | Maturation | t/Year | 24 (2024) to 1,082 (1996) |
| | Natural death | t/Year | 12 (2024) to 359 (1997) |
| | Catch | t/Year | 0 (2024–2027) to 718 (1997) |

| Group | Variable | Unit | Value range (year) |
|---|---|---|---|
| Flows young whitefish | Recruitment | t/Year | 35 (2023) to 1,933 (1987) |
| | Young mortality | t/Year | 18 (2024) to 812 (1996) |
| Flows zooplankton | Zooplankton growth | kInd/m²/Year | 686 (2025) to 2,747 (1977) |
| | Zooplankton mortality | kInd/m²/Year | 569 (2050) to 2,181 (1977) |
| | Grazing by whitefish | kInd/m²/Year | 10 (2024) to 726 (1986) |
| | Grazing by sticklebacks | kInd/m²/Year | 0 (before 2012) to 78 (2018) |
| Effect factors | Food effect | Dmnl | 0.98 (2050) to 1.43 (1977) |
| | Oxygen effect | Dmnl | 0.71 (1973) to 1.07 (1950) |
| | Stickleback egg effect | Dmnl | 0.49 (2012–2023) to 1.0 (before 2012) |
| | Stickleback pressure | Dmnl | 0 (before 2012) / 1 (2012–2023) / 0.2 (from 2024) |
| | Model survey index | Dmnl | 0.76 (2024) to 22.5 (1997) |

*Table 5: Outputs of the model*

## Model Analysis

### Numerical stability

To analyse numerical stability, various simulations were performed using different time steps and integration methods. Simulations were conducted using time steps of 0.0625, 0.125, 0.25, 0.5, and 1, as well as the Euler and Runge-Kutta 4 integration methods, which can be seen in Figure 5.

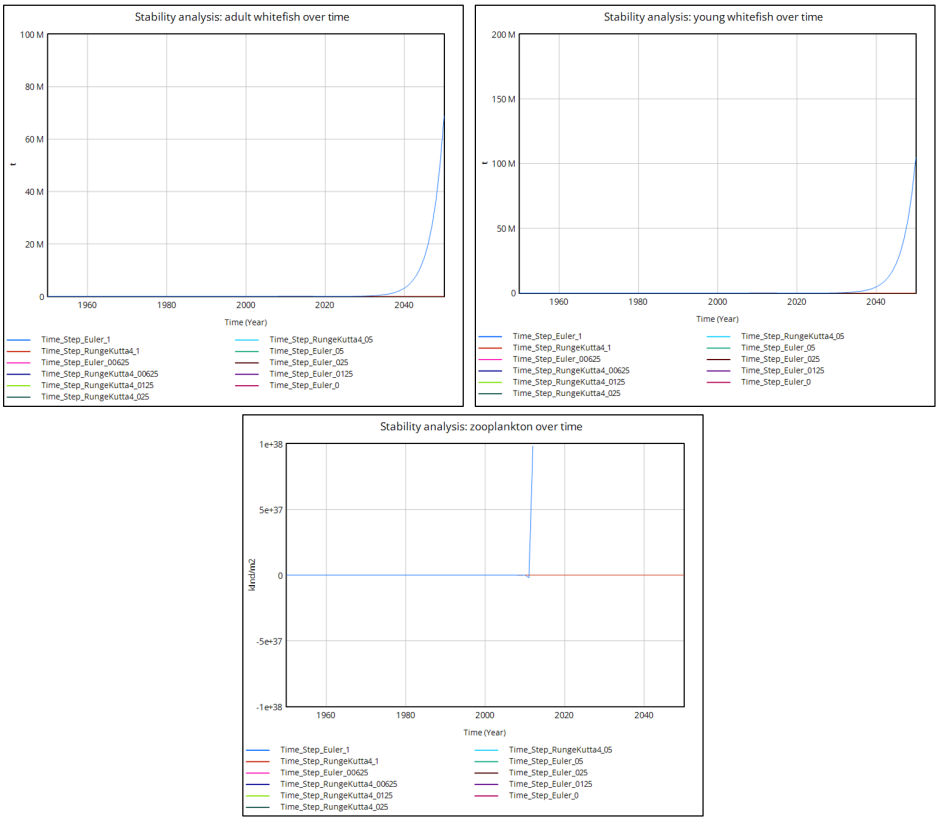



*Figure 5: Stability analysis*

Since the simulation using the Euler integration method and a time step of 1 is unstable, as shown in Figure 5, it was not included in the more detailed analysis. Figure 6 shows that the

historical phase up to approximately 2022 is numerically robust, with a very small deviation of less than 1%. The scenario up to 2050 is not numerically stable and depends heavily on the time step. All results in the following sections were computed with the Euler integration method and a time step of 0.0625 years.

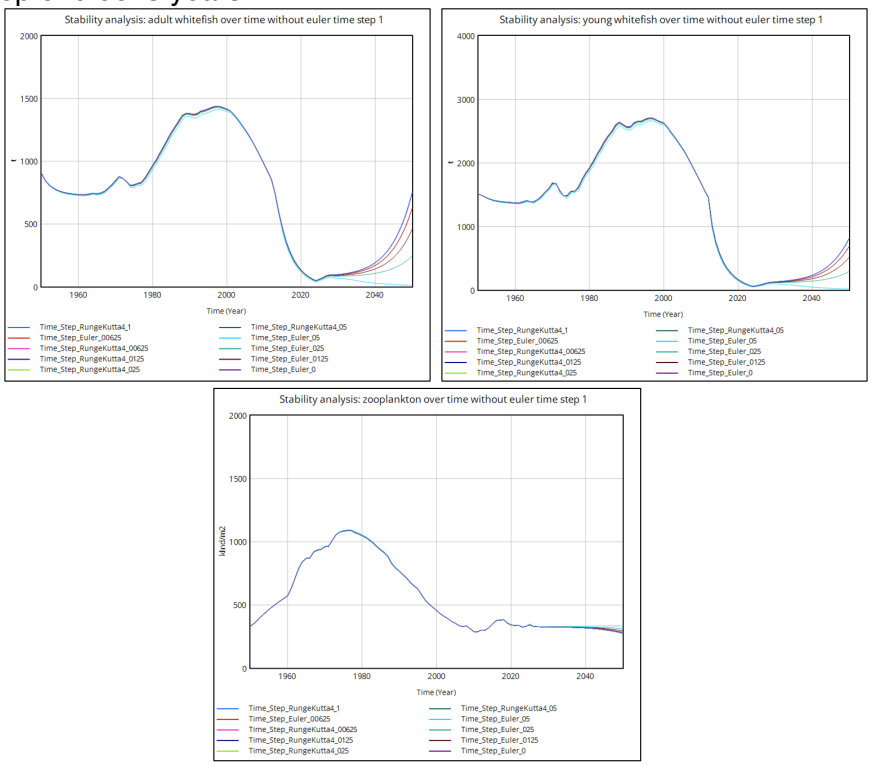



*Figure 6: Stability analysis without Euler integration (time step 1)*

## Degree of validity

The model's degree of validity was assessed by examining its units and checking the syntax for unlinked variables. The unit testing feature in Vensim was used to test the units, a screenshot of it can be seen in Figure 7. In addition, the parameters were checked for plausibility by comparing them with the literature (see Literature, p.22). The model clearly distinguishes between the assumptions and the calibrations.

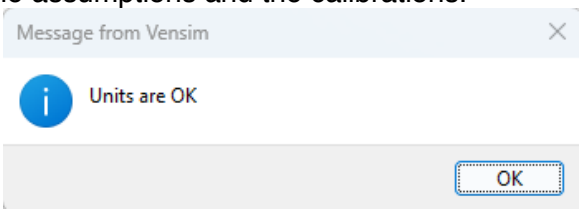



*Figure 7: Vensim unit check*

## Sensitivity

To test the sensitivity of the model, eight parameters (maximum recruitment rate, adult mortality rate, young mortality rate, maturation time, stickleback egg impact, stickleback grazing rate, maximum zooplankton production, and food half-saturation) were varied individually by ±10%, while all other parameters remained constant. Figure 8 shows that the qualitative behavior of the model remains consistent across all variants: The adult whitefish population increases until around 1990, then drops to a minimum by around 2024, and subsequently recovers. However, the magnitude of the response differs significantly between the parameters, with variations in the maximum recruitment rate and food half saturation causing the greatest deviations from the baseline run. Even in the historical phase, stock levels vary noticeably (peaking around 1997 at between approximately 400 and 3,500 t, depending on the variant), as the calibrated parameters were altered without recalibration. The variation is particularly large in the projection through 2050, where values range from nearly zero to over

6,000 t, depending on the parameter selection. The recovery of the whitefish stock after 2024 is thus highly dependent on the uncertainty of individual parameters, which is why the results for 2050 should be interpreted as scenarios rather than forecasts.

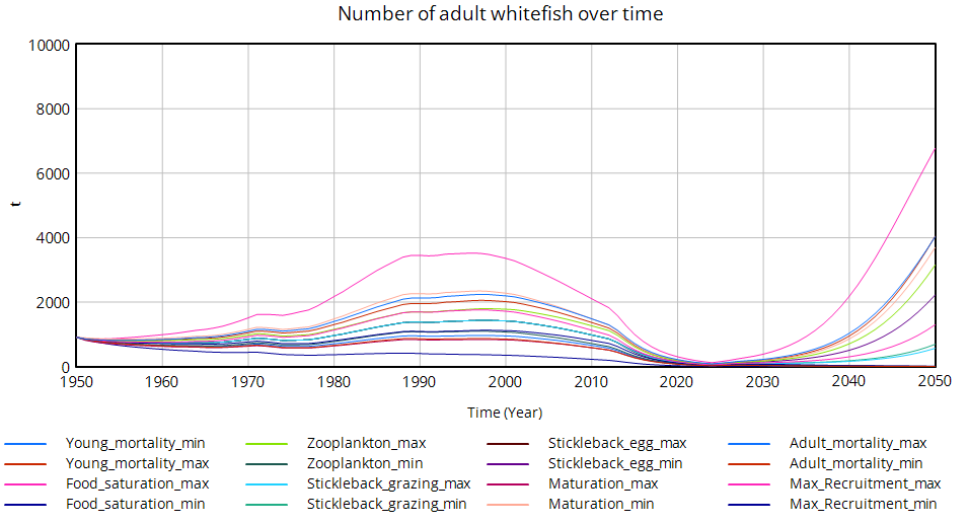



*Figure 8: Sensitivity analysis*

# Results

## General model behaviour

The populations of young and adult whitefish follow nearly parallel trends, as shown in Figure 9. This is plausible, since adult whitefish produce the young, which grow into adult whitefish after about 2.5 years. It should be noted that the two axes in the graph on the left have different scales and that the population of young whitefish is approximately twice as large as that of adult whitefish. The adult whitefish population initially declined from approximately 900 t in 1950 to about 730 t around 1960, before rising to a peak of approximately 1,430 t by 1997. Starting in 2000, a sharp decline is evident, reaching its lowest point in 2024/2025 at only about 50 t. After that, the population slowly recovers and rises again to approximately 620 t by 2050. The young whitefish population also shows a sharp decline, followed by a recovery to about 700 t.

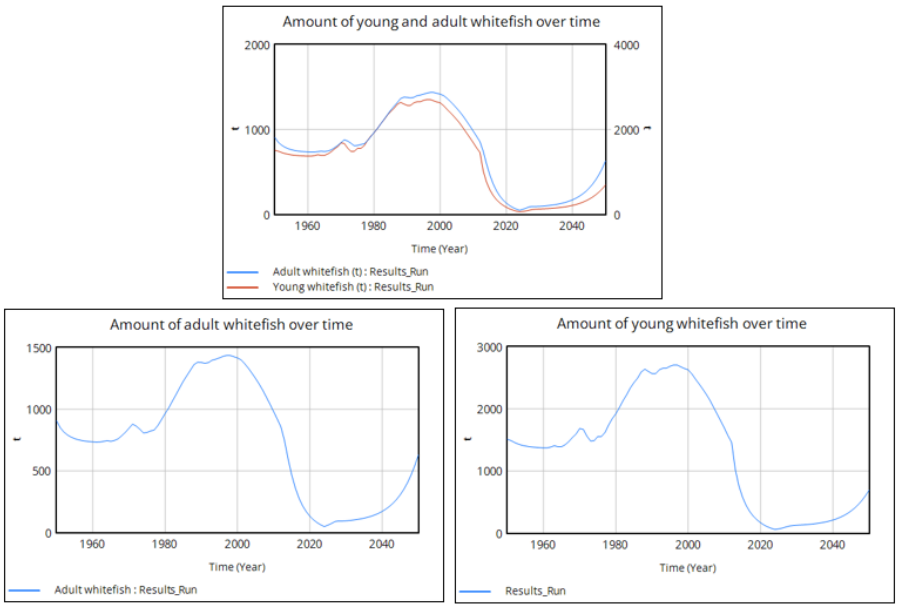



*Figure 9: Development of the population over time*

The influencing factors of food, oxygen, and sticklebacks are shown in Figure 10 and have varying effects on whitefish recruitment. The food effect rises to a maximum of approximately 1.43 by about 1975–1980 and then declines to about 1.0 by 2010. Food thus initially promotes recruitment, while its influence is largely neutral later. The oxygen effect is mostly close to 1.0 but shows occasional declines to approximately 0.7–0.8 in the 1960s and 1970s. From 2000 onward, the oxygen effect remains virtually constant. The stickleback egg effect drops to about 0.49 starting in 2012 and rises again to approximately 0.90 starting in 2024. The decline in the whitefish population in the model can be explained by the food effect and after 2012 also by the stickleback egg effect.

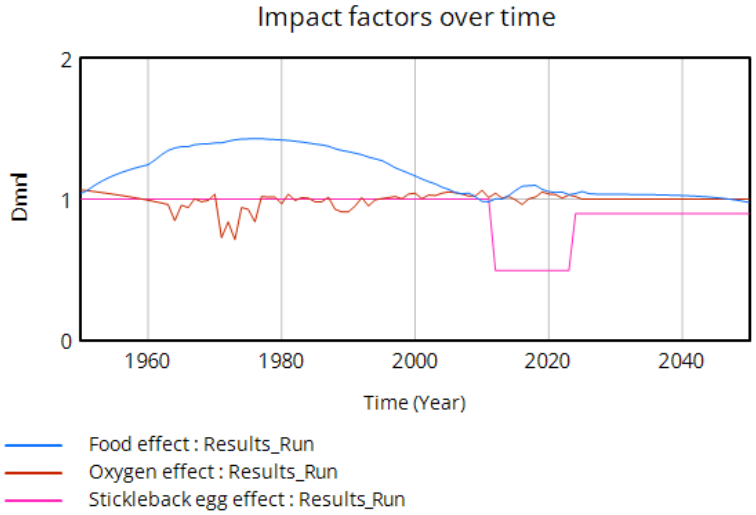



*Figure 10: Impact factors over time*

The inflows and outflows of the adult population shown in Figure 11 follow a similar trend to the population's own development (see Figure 9). The flows increase until around 1990–2000, before declining sharply through 2024. Maturation reaches a maximum of approximately 1,050 t per year. The main outflows are natural death and catch. In 1997, the stock is nearly in equilibrium, as maturation roughly equals the combined total of natural death and catch. Starting in 2025, maturation exceeds outflows, allowing the population to recover in the long term.

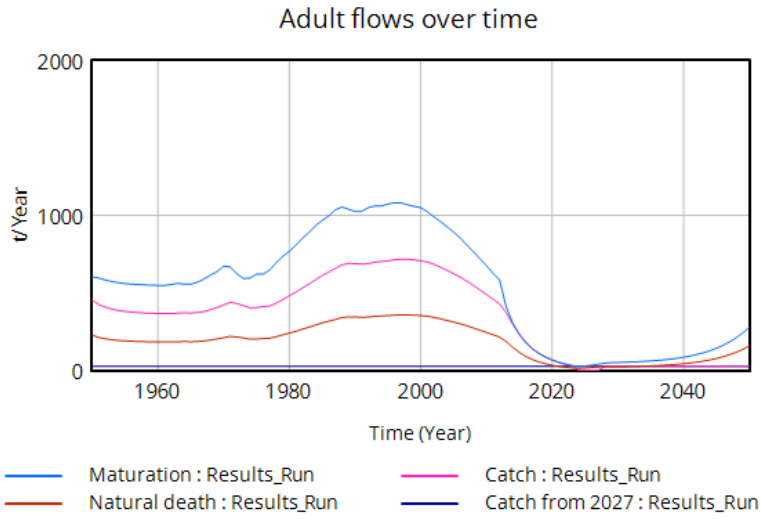



*Figure 11: Adult flows over time*

## Reproducing the historical data

The first research question asks whether the model can reproduce the observed development of Lake Constance between 1950 and 2025. Therefore, the following hypothesis was tested: The model can reproduce the historical data.

The model and the observational data on zooplankton generally show a similar long-term trend, as can be seen in Figure 12. The modelled population rises from approximately 330 kInd/m² to a maximum of about 1,080 around 1975–1980 and then declines again to about 330 by 2008. A slight increase is then evident before the population stabilizes at around 300. The observational data, on the other hand, show stronger annual fluctuations and reach a maximum of approximately 1,470 kInd/m² around 1970. After that, the population declines sharply and, starting in 1997, fluctuates primarily between about 200 and 450 kInd/m². The model reflects the long-term trend but underestimates the peak values and cannot represent the annual fluctuations. It is also important to note that the constant values before 1961 and after 2025 result from the lookup function and do not represent actual measured values.

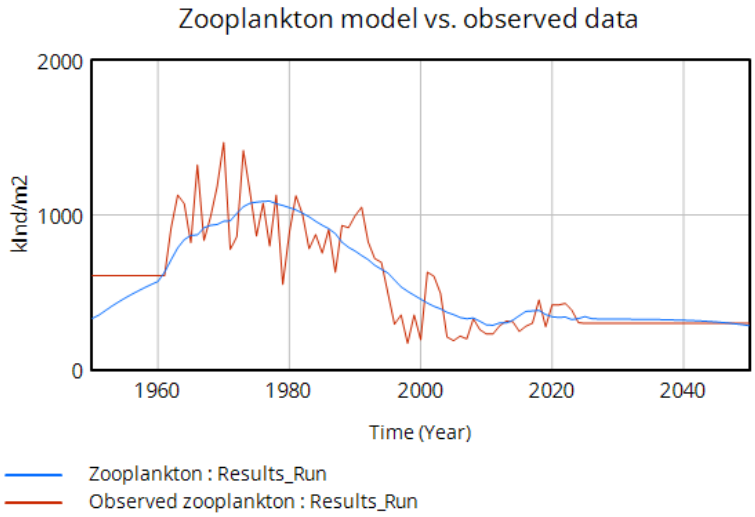



*Figure 12: Zooplankton model vs. observed data*

The modelled survey index is shown in Figure 13 and exhibits a long-term trend like that of the observed data. The model reaches a maximum of approximately 22 around 1997 and then declines steadily to about 1 in 2024. Subsequently, a rise back to about 10 is projected by 2050. The observed data initially fluctuate between approximately 15 and 33 before the index drops sharply starting in 2013 and remains at a low level. The model generally replicates this decline, though less abruptly and with a time lag. It is important to note that the constant values prior to 1997 do not represent actual measured values but are generated by the lookup function.

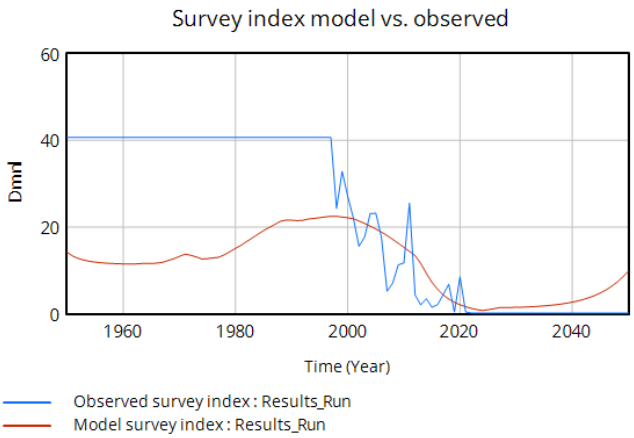



*Figure 13: Survey index model vs. observed*

The modelled catch levels in Figure 14 generally follow a similar trend to the observed data but exhibit less fluctuation. Initially, the modelled catch is approximately 450 t and reaches a peak of about 720 t around 1998. It then declines to nearly zero by 2024, before rising to 25 t per year following the fishing ban. The observed data, by contrast, show pronounced peaks of approximately 1,200 t in 1977 and around 1,000 t in 1990. The model also reflects the long-term decline beginning in 2000. The discrepancies can be explained, among other things, by the fact that the catch rate remains constant in the model, whereas actual catch also depends on fishing effort.

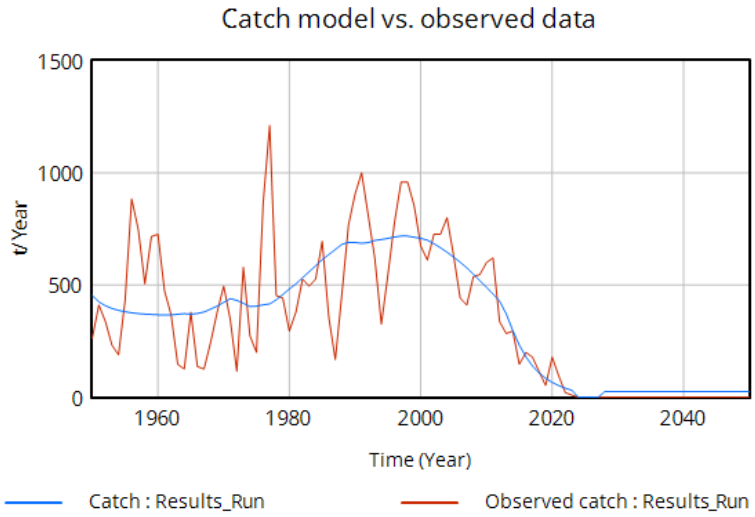



*Figure 14: Catch model vs. observed data*

The model reproduces the long-term pattern and the timing of the main changes, namely the rise until the 1990s and the collapse after 2000. It does not reproduce year-to-year variability or peak magnitudes, and fit quality decreases from zooplankton to survey index to catch. Two limits apply. The parameters were calibrated on the same data, so this is not an independent test. In addition, the results are numerically robust up to about 2022 for time steps ≤ 0.0625 years (differences below 1 %), whereas projections to 2050 depend on the time step.

## Fishing ban and decline of stickleback population contribution to the whitefish stock recovery

Because the fishing ban and the stickleback decline both started in 2024, their effects cannot be separated using observational data. The second research question was therefore addressed with the following hypothesis: The model can show the contribution effects of the 2024 fishing ban and the decline of the stickleback population to the recovery of the whitefish stock.

To isolate the two effects, four scenarios were simulated (2×2 design): ban and decline (baseline), ban only, decline only, and neither. “No decline” means that the stickleback pressure remains at 1 after 2024; “no ban” means that fishing continues at the normal level until May 2027. Starting in May 2027, a catch limit of 25 t per year applies in all scenarios; additionally, all scenarios were calculated with zero catch to isolate the pure effects.

Figure 15 shows the four scenarios with the planned catch of 25 t/year. All runs are identical until 2024 and decline from about 135 t in 2020 to about 50 t. After 2024 they separate. Only the base scenario recovers: it rises to about 90 t by 2027, grows only slowly until about 2030 (95 t) and then increasingly faster to 620 t in 2050. The other three scenarios collapse. With the ban only, the stock still rises to about 65 t by 2027, but falls below 30 t by 2030, because the catch resumes while recruitment remains halved. With the stickleback decline only or with neither measure, the stock declines towards zero (below 5 t in 2050). Therefore, in the scenario with a catch of 25 t, neither measure is sufficient on its own, but they work synergistically.

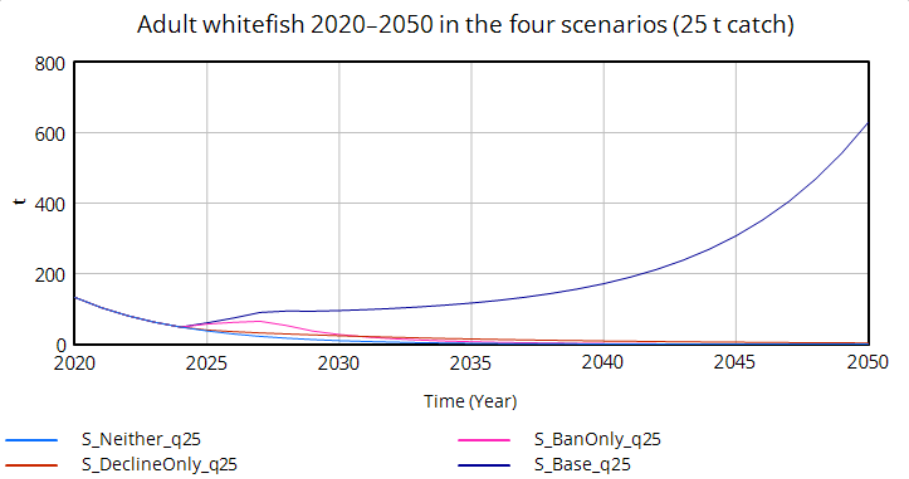



*Figure 15: Adult whitefish 2020-2025 in the four scenarios (25t catch)*

To isolate the pure effects, Figure 16 shows the same scenarios without any catch after 2027. Here all runs grow, but at very different rates. The base scenario reaches about 3,779 t in 2050 and the decline-only scenario about 2,000 t, whereas the ban-only (about 145 t) and neither scenario (about 55 t) stays close to the axis. The curves are hardly distinguishable until about 2030 and diverge strongly only afterwards, because higher recruitment accumulates in the stock. The absolute values are unrealistic, as they exceed the historical maximum of about 1,430 t. The model has no density dependence for adults apart from the zooplankton feedback, so these values are used only for relative comparison.

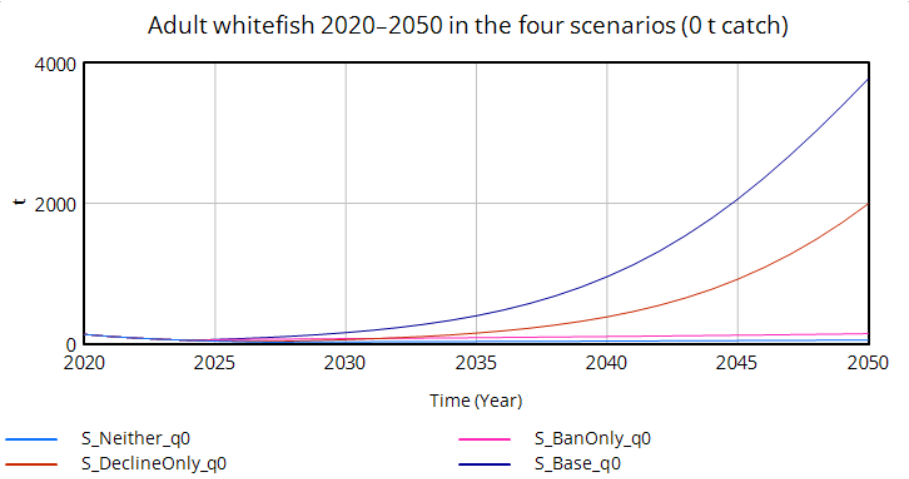



*Figure 16: Adult whitefish 2020-2050 in the four scenarios (0t catch)*

The ban was the main driver of the short-term increase (about three quarters by 2027). The stickleback decline is the main driver of the long-term recovery (about three quarters by 2050 without catch, Figure 16). Under the planned catch, the recovery requires both (Figure 15).

## Calculation of the maximum annual catch from 2027 onwards to keep the stock above a reference level (50t) until 2050

Since fishing will resume in May 2027 in Lake Constance, the third hypothesis examines how much whitefish can be caught without endangering the recovery of the stock: The model can simulate the maximum annual catch that can be permitted from 2027 onwards to ensure that the stock remains above a reference level until 2050, under both low and renewed stickleback pressure.

The reference level was defined as 50 t, which corresponds to the 2024 minimum. The adult stock must not fall below this level between May 2027 and 2050. The catch from 2027 was varied under low (0.2, base, assumption) and renewed (1.0) stickleback pressure.

Under low stickleback pressure, see Figure 17, all runs start at about 90 t in 2027 and then separate according to the catch. The lower the catch, the stronger the growth: without catch the stock reaches about 3,780 t in 2050, with 10 t/year about 2,850 t, with 20 t/year about 1,540 t and with 25 t/year about 630 t. At 30 t/year, however, the stock no longer grows and falls to about 14 t. There is thus a clear threshold between 25 and 30 t/year, and the maximum catch that keeps the stock above the reference level is about 27 t/year.

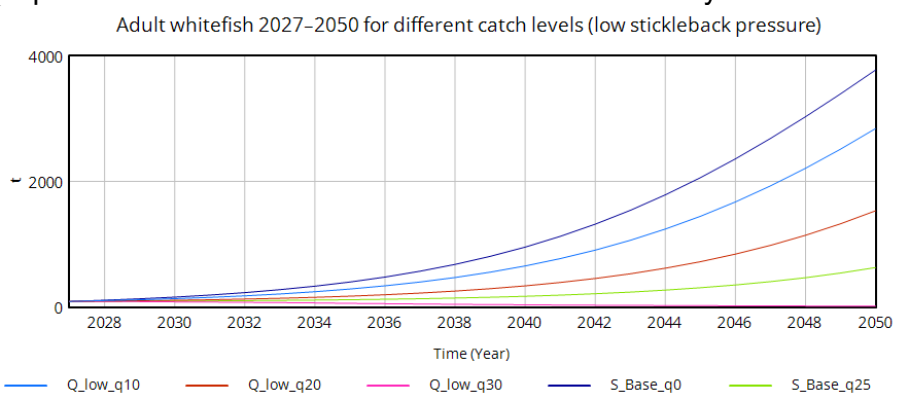



*Figure 17: Adult whitefish 2027-2050 for different catch levels (low stickleback pressure)*

Under renewed stickleback pressure, see Figure 18, the stock also starts at about 90 t, but the recovery is much weaker. Without catch it rises only to about 220 t in 2050. A catch of 5 t/year keeps the stock almost constant at about 100 t, which is close to the sustainable level. With 7 t/year the stock declines slowly to about 55 t in 2050 and would continue to fall afterwards, so this catch meets the 50 t criterion only because the simulation ends in 2050. With 10 and 15 t/year the stock falls to about 4 t and almost zero, respectively. The maximum catch is therefore only about 5–7 t/year. Even without catch, the stock reaches only about 220 t in 2050.

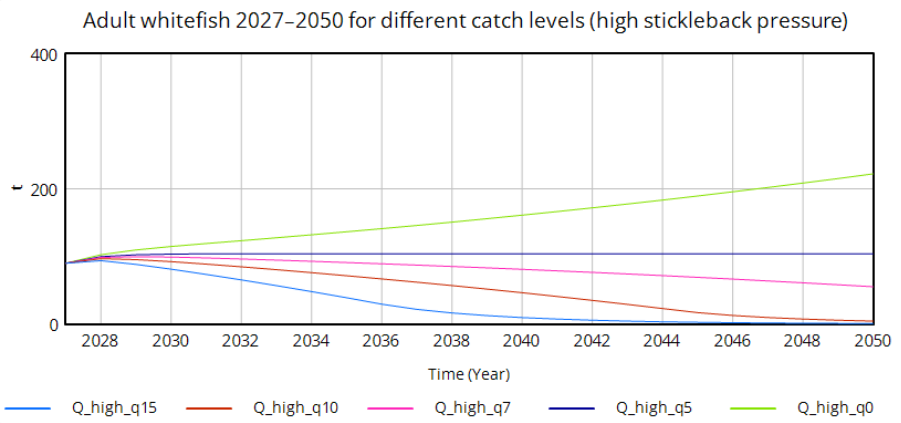



*Figure 18: Adult whitefish 2027-2050 for different catch levels (high stickleback pressure)*

Under low stickleback pressure, a catch of up to about 27 t/year can be permitted, and under renewed pressure only about 5–7 t/years above these values, the model predicts a gradual decline.

# Discussion

## Open issues

**Water temperature and climate change:** Water temperature is not explicitly included in the model, and the projections to 2050 do not incorporate climate-change scenarios; historical phosphorus and deep-water oxygen measurements are used as external inputs rather than being predicted dynamically. Temperature could influence recovery through egg development, larval growth, food availability and deep-water oxygen replenishment. Modest historical warming has sometimes been associated with stronger whitefish recruitment, whereas experiments with Upper Lake Constance whitefish indicate that further warming can impair early-life survival and performance. The net effect on recovery therefore cannot be determined from this model, but the uncertainty should not be interpreted as evidence that beneficial and harmful outcomes are equally likely.

**Other ecological drivers:** The model omits quagga mussels, stocking, cormorants, seasonal dynamics and detailed whitefish age and size structure. Quagga mussels are particularly relevant because their filtration removes phytoplankton and can divert nutrients and energy away from the open-water food web. They may therefore reduce the resources supporting zooplankton and indirectly limit the food available to pelagic whitefish; describing this as indirect food competition is more accurate than implying that mussels and whitefish necessarily eat the same prey. Research has documented food-web changes following the invasion, but the magnitude of the resulting effect on whitefish recovery remains (Eawag, 2022).

**Spatial scope of the data:** The model explicitly represents Upper Lake Constance, and its principal fishery calibration data are from that basin. Its results should consequently be interpreted within that spatial scope rather than automatically extended to Lower Lake Constance or every habitat within the lake. Here, “Upper Lake Constance” means the Obersee basin, not the upper layers of the water column.

**Interpretation:** The model is a tool for comparing scenarios under selected drivers and assumptions, not a complete representation of the ecosystem. Its recovery trajectories and catch thresholds should therefore be treated as conditional scenario results rather than reliable forecasts or independently validated catch recommendations.

## Evidence and implications

**Temperature effects are conditional:** This report treats oxygen and phosphorus as prescribed inputs and excludes water temperature from its system boundaries; this does not mean historical climate effects are entirely absent, because measured environmental inputs may already reflect some past climatic influence. The limitation is that the model does not explicitly simulate temperature-dependent biological rates or future climate-driven changes in those inputs.

Eckmann’s 2013 review describes benefits associated with modest historical warming, including faster egg development and improved early growth, while also discussing risks from impaired winter mixing. Roberts, Chucholl and Brinker’s 2024 experiments specifically address Upper Lake Constance whitefish and support concern about adverse effects of further warming. These findings concern different temperature ranges and mechanisms, so they do not justify a simple assertion that warming always helps or always harms every life stage (Eckmann, 2013) (Roberts et al., 2024).

**Quagga effects are indirect:** The relevant pathway is phytoplankton filtration and altered nutrient routing, with possible consequences for zooplankton and the fish that depend on it. Baer and colleagues’ 2022 study found altered energy pathways after the quagga invasion

and reported that pelagic whitefish still fed on zooplankton, whereas some benthic fish consumed mussels. Consequently, the limitation is an omitted, changing food-web mechanism rather than a demonstrated, quantified reduction in whitefish recovery attributable to mussels alone (Eawag, 2022).

**Additional methodological caveats:** The numerical sensitivity in the projections to 2050, the substantial sensitivity to biological parameters, and the lack of independent validation because historical observations were also used for calibration, present the limitations of this model. These limitations are directly relevant to interpreting the estimated catch thresholds: the reported values are conditional on the model setup and should not be presented as established sustainable harvest limits.

There is also the concern of a possibly unrealistic stock growth in some zero-catch scenarios and limited density dependence beyond the zooplankton feedback. This reinforces the distinction between exploring mechanisms within the model and demonstrating how the real ecosystem will develop.

## Learning from the model

Reproduction of historical data (hypotheses 1): The results show that the model can reproduce the broad historical trends in zooplankton and whitefish development, including the increase in whitefish abundance until the 1990s and the subsequent decline. However, it does not capture the observed annual fluctuations or the magnitude of some peaks. The agreement also differs between the datasets, with the closest fit for zooplankton and larger deviations for the survey index and catch data. Therefore, the hypothesis that the model can reproduce the historical data is only partially supported. Furthermore, since all three datasets were used for calibration, this agreement should be interpreted as an in-sample fit rather than independent evidence of predictive accuracy.

Whitefish recovery (hypotheses 2&3): The experiments indicate that whitefish recovery depends on the interaction between protecting adults and improving recruitment, rather than on either factor alone. The fishing ban provides the greater short-term benefit by immediately reducing adult mortality, whereas reduced stickleback pressure supports longer-term recovery as additional offspring mature and reproduce. Under the simulated catch of 25 tonnes per year after reopening, only the combination of both changes produces recovery, showing that an initial stock increase does not necessarily indicate sufficient capacity to support renewed fishing. The catch experiments also demonstrate that a fixed quota becomes increasingly demanding as the stock declines, because it removes a larger proportion of the remaining adults. Although the reported catch thresholds differ substantially between low and renewed stickleback pressure, they only meet the selected requirement of keeping adult biomass above 50 tonnes until 2050 and should not be interpreted as established sustainable yields.

In relation to its purpose, the model helps explain how reproduction, food availability, maturation delays and fishing can generate contrasting recovery trajectories within a simplified representation of Upper Lake Constance. Its reproduction of broad historical trends supports its usefulness for exploring these mechanisms, but does not independently validate them because the same observations were used for calibration. The model boundary also determines which questions can be answered: prescribing stickleback pressure allows its consequences to be compared, but cannot explain the population’s decline or predict its return. Furthermore, strong parameter sensitivity, time-step-dependent projections and unrealistic growth in some zero-catch scenarios limit confidence in exact biomass and catch estimates. The main learning is therefore about interacting pressures and conditional management outcomes, rather than a precise prediction of the lake’s future development.

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### Figures

Figure 1: Model Overview ..... 6
Figure 2: Whitefish life cycle ..... 6
Figure 3: Zooplankton food web ..... 7
Figure 4: Stickleback pressure and fishing outputs ..... 7
Figure 5: Stability analysis ..... 12
Figure 6: Stability analysis without Euler integration (time step 1) ..... 13
Figure 7: Vensim unit check ..... 13
Figure 8: Sensitivity analysis ..... 14
Figure 9: Development of the population over time ..... 14
Figure 10: Impact factors over time ..... 15
Figure 11: Adult flows over time ..... 15
Figure 12: Zooplankton model vs. observed data ..... 16
Figure 13: Survey index model vs. observed ..... 16
Figure 14: Catch model vs. observed data ..... 17
Figure 15: Adult whitefish 2020-2025 in the four scenarios (25t catch) ..... 18
Figure 16: Adult whitefish 2020-2050 in the four scenarios (0t catch) ..... 18
Figure 17: Adult whitefish 2027-2050 for different catch levels (low stickleback pressure) ..... 19
Figure 18: Adult whitefish 2027-2050 for different catch levels (high stickleback pressure) ..... 19
Figure 19: Vensim Model ..... 25

### Tables

Table 1: State variables of the model ..... 8
Table 2: Auxiliary variables of the model ..... 10
Table 3: ODE's of the model ..... 10
Table 4: Calibrated parameters used in the model ..... 11
Table 5: Outputs of the model ..... 12

# Appendix

### Link to GitHub Repository

GitHub Repository: https://github.com/the-nerd-sloth/co1-semester-project

## Model as Image

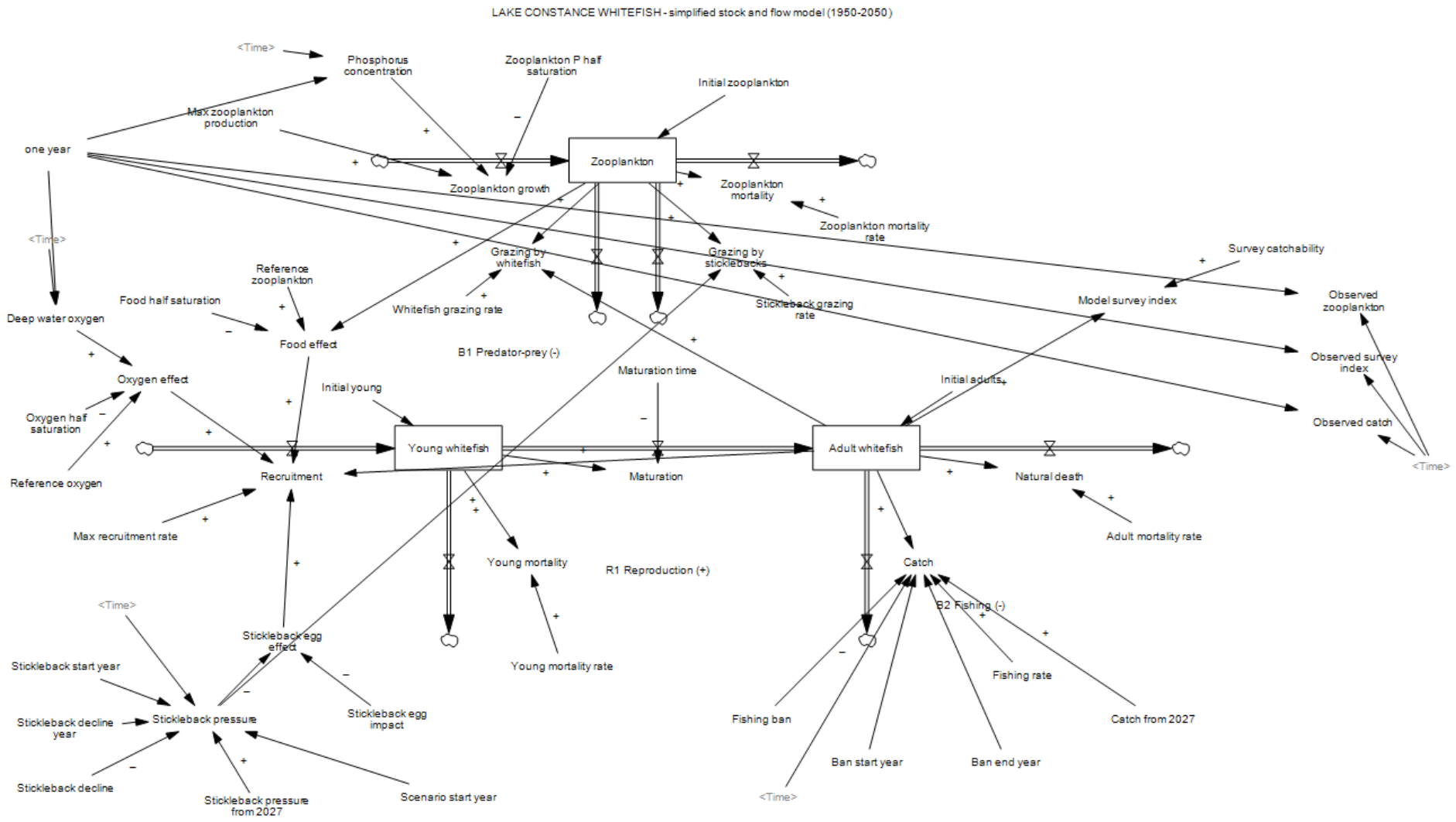



*Figure 19: Vensim Model*

## Model as Text

Adult mortality rate
Units : 1/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION: natural mortality of adults.
Equations : Adult mortality rate = 0.25
Use : Natural death
Adult whitefish
Units : t
Variable type : Stock
Group : .whitefish model (1)
Comment : STOCK: adult (catchable, spawning) whitefish biomass in Lake Constance (Obersee).
Equations : Adult whitefish =
INTEG( Maturation
- Natural death
- Catch ,
Initial adults )
Cause : Catch
Cause : Initial adults
Cause : Maturation
Cause : Natural death
Use : Catch
Use : Grazing by whitefish
Use : Model survey index
Use : Natural death
Use : Recruitment
Ban end year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Fishing reopens in May 2027 (SRF, 2026).
Equations : Ban end year = 2027.33
Use : Catch
Ban start year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Fishing ban from January 2024 (IBKF; Frei & Kugler, 2026).
Equations : Ban start year = 2024
Use : Catch
Catch
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Catch (B2). Before 2024: a fixed share of the adult stock. 2024 - April 2027: fishing ban. From May 2027: a fixed annual catch (scenario), at most Fishing rate x stock.
Equations : Catch =
IF THEN ELSE ( Time
< Ban start year ,
Fishing rate
* Adult whitefish ,
IF THEN ELSE ( Time
< Ban end year ,

( 1
- Fishing ban )
* Fishing rate
* Adult whitefish ,
MIN ( Catch from 2027 ,
Fishing rate
* Adult whitefish ) ) )
Cause : Time
Cause : Adult whitefish
Cause : Ban end year
Cause : Ban start year
Cause : Catch from 2027
Cause : Fishing ban
Cause : Fishing rate
Use : Adult whitefish
Catch from 2027
Units : t/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : SCENARIO: annual catch from May 2027. Target of the IBKF: 20-30 t (SRF, 2026).
Equations : Catch from 2027 = 25
Use : Catch
Deep water oxygen
Units : mgO2/L
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : DATA: minimum oxygen 1 m above the lake bottom (IGKB) 1963-2025. 1950 = 8 mgO2/L is an ASSUMPTION. Held at the 2025 value afterwards.
Equations : Deep water oxygen =
WITH LOOKUP( Time
/ one year ,
((1950,8),(1963,5.2),(1964,3.5),(1965,5.1),(1966,4.8),(1967,6.1),(1968,5.6),(1969,5.8),(1970,7),(1971,2.4),(1972,3.4),(1973,2.3),(1974,4.8),(1975,4.6),(1976,3.4),(1977,6.5),(1978,6.4),(1979,6.4),(1980,5.3),(1981,7),(1982,5.8),(1983,6.3),(1984,6.2),(1985,5.6),(1986,5.6),(1987,6.36),(1988,4.6),(1989,4.3),(1990,4.3),(1991,5),(1992,6.3),(1993,5),(1994,5.9),(1995,6.12),(1996,6.3),(1997,6.5),(1998,6.1),(1999,7),(2000,7.2),(2001,6.1),(2002,6.8),(2003,6.7),(2004,7.2),(2005,7.5),(2006,7.3),(2007,7),(2008,6.6),(2009,6.6),(2010,7.9),(2011,6.3),(2012,7.2),(2013,6.2),(2014,6.5),(2015,5.9),(2016,5.2),(2017,6.1),(2018,6.4),(2019,7.5),(2020,7),(2021,6.9),(2022,6.2),(2023,6.8),(2024,6.5),(2025,6) )
)
Cause : Time
Cause : one year
Use : Oxygen effect
FINAL TIME
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Simulation end.
Equations : FINAL TIME = 2050
Fishing ban
Units : Dmnl
Variable type : Constant
Group : .whitefish model (1)
Comment : SCENARIO switch: 1 = ban 2024-2027 as decided, 0 = no ban (fishing continues).
Equations : Fishing ban = 1
Use : Catch

Fishing rate
Units : 1/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION: share of the adult stock caught per year before the ban (about 40 % per year).
Equations : Fishing rate = 0.5
Use : Catch
Food effect
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : More zooplankton -> better fed whitefish -> more offspring survive. 1 at the reference zooplankton level.
Equations : Food effect =
( Zooplankton
/ ( Zooplankton
+ Food half saturation ) )
/ ( Reference zooplankton
/ ( Reference zooplankton
+ Food half saturation ) )
Cause : Zooplankton
Cause : Food half saturation
Cause : Reference zooplankton
Use : Recruitment
Food half saturation
Units : kInd/m2
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: zooplankton level at which the food effect is halved.
Equations : Food half saturation = 213.009
Use : Food effect
Grazing by sticklebacks
Units : kInd/(Year*m2)
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Zooplankton eaten by sticklebacks: competition with whitefish for food.
Equations : Grazing by sticklebacks =
Zooplankton
* Stickleback grazing rate
* Stickleback pressure
Cause : Zooplankton
Cause : Stickleback grazing rate
Cause : Stickleback pressure
Use : Zooplankton
Grazing by whitefish
Units : kInd/(Year*m2)
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Zooplankton eaten by whitefish: more whitefish -> less food left (B1 predator-prey).
Equations : Grazing by whitefish =
Zooplankton
* Whitefish grazing rate
* Adult whitefish
Cause : Adult whitefish
Cause : Zooplankton

Cause : Whitefish grazing rate
Use : Zooplankton
Initial adults
Units : t
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: adult stock in 1950.
Equations : Initial adults = 906.673
Use : Adult whitefish
INITIAL TIME
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Simulation start.
Equations : INITIAL TIME = 1950
Use : Time
Initial young
Units : t
Variable type : Constant
Group : .whitefish model (1)
Comment : Young stock in 1950, in equilibrium with the 1950 conditions.
Equations : Initial young = 1511.98
Use : Young whitefish
Initial zooplankton
Units : kInd/m2
Variable type : Constant
Group : .whitefish model (1)
Comment : Zooplankton in 1950, in equilibrium with the 1950 conditions.
Equations : Initial zooplankton = 331.731
Use : Zooplankton
Maturation
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Young fish become adults after about 2.5 years.
Equations : Maturation =
Young whitefish
/ Maturation time
Cause : Young whitefish
Cause : Maturation time
Use : Adult whitefish
Use : Young whitefish
Maturation time
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION: whitefish mature at about 2-3 years.
Equations : Maturation time = 2.5
Use : Maturation
Max recruitment rate
Units : 1/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: offspring (t) per t of adults and year at reference food and oxygen.
Equations : Max recruitment rate = 1.05091
Use : Recruitment

Max zooplankton production
Units : kInd/(Year*m2)
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED against observed zooplankton 1961-2025.
Equations : Max zooplankton production = 3444.7
Use : Zooplankton growth
Model survey index
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Model value comparable to the observed survey index.
Equations : Model survey index =
Adult whitefish
* Survey catchability
Cause : Adult whitefish
Cause : Survey catchability
Natural death
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Natural death of adults.
Equations : Natural death =
Adult whitefish
* Adult mortality rate
Cause : Adult whitefish
Cause : Adult mortality rate
Use : Adult whitefish
Observed catch
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : DATA for comparison: whitefish catch of professional fishers, Obersee (IGKB, Abb. 1, read from the chart); 2023 = 9.8 t (Frei & Kugler, 2026); 2024-2025 fishing ban.
Equations : Observed catch =
WITH LOOKUP( Time
/ one year ,
((1950,263),(1951,411),(1952,337),(1953,232),(1954,189),(1955,432),(1956,884),(1957,747),(1958,505),(1959,716),(1960,726),(1961,474),(1962,368),(1963,147),(1964,126),(1965,379),(1966,137),(1967,126),(1968,242),(1969,379),(1970,495),(1971,347),(1972,116),(1973,579),(1974,274),(1975,200),(1976,863),(1977,1211),(1978,453),(1979,442),(1980,295),(1981,379),(1982,526),(1983,495),(1984,526),(1985,695),(1986,358),(1987,168),(1988,463),(1989,768),(1990,905),(1991,1000),(1992,811),(1993,621),(1994,326),(1995,547),(1996,779),(1997,958),(1998,958),(1999,853),(2000,674),(2001,611),(2002,726),(2003,726),(2004,800),(2005,632),(2006,442),(2007,411),(2008,537),(2009,547),(2010,600),(2011,621),(2012,337),(2013,284),(2014,295),(2015,147),(2016,200),(2017,179),(2018,116),(2019,53),(2020,179),(2021,95),(2022,21),(2023,9.8),(2024,0),(2025,0) )
)
Cause : Time
Cause : one year
Observed survey index
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : DATA for comparison: whitefish survey index (IGKB/IBKF) 1997-2022.
Equations : Observed survey index =

WITH LOOKUP( Time
/ one year ,
((1997,40.66),(1998,24.29),(1999,32.86),(2000,27.11),(2001,22.09),(2002,15.58),(2003,17.7)
,(2004,23.13),(2005,23.19),(2006,17.71),(2007,5.21),(2008,7.06),(2009,11.29),(2010,11.72),(
2011,25.52),(2012,4.34),(2013,2.07),(2014,3.49),(2015,1.52),(2016,2.11),(2017,4.42),(2018,
6.82),(2019,0.43),(2020,8.51),(2021,0.42),(2022,0.19) )
)
Cause : Time
Cause : one year
Observed zooplankton
Units : kInd/m2
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : DATA for comparison: crustacean zooplankton (IGKB/BOWIS) 1961-2025, in 1000 individuals per m2.
Equations : Observed zooplankton =
WITH LOOKUP( Time
/ one year ,
((1961,607.893),(1962,915.8),(1963,1129.5),(1964,1074.1),(1965,821.4),(1966,1325.1),(196
7,835.4),(1968,986.8),(1969,1184.5),(1970,1470.2),(1971,778),(1972,859.8),(1973,1418),(19
74,1152.11),(1975,864.5),(1976,1077.1),(1977,799.1),(1978,1129.9),(1979,551.08),(1980,88
5.43),(1981,1125.05),(1982,1004.3),(1983,782),(1984,873.9),(1985,754.36),(1986,907.5),(19
87,629.819),(1988,932.6),(1989,917.976),(1990,995.625),(1991,1051.25),(1992,825.317),(1
993,719.284),(1994,694.464),(1996,294.058),(1997,353.985),(1998,169.656),(1999,353.945)
,(2000,193.013),(2001,630.581),(2002,604.81),(2003,494.407),(2004,211.443),(2005,187.55
7),(2006,217.555),(2007,199.061),(2008,328.815),(2009,258.681),(2010,230.509),(2011,231
.258),(2012,283.287),(2013,314.665),(2014,310.793),(2015,247.703),(2016,281.155),(2017,
300.725),(2018,450.999),(2019,277.512),(2020,420.152),(2021,418.637),(2022,428.761),(20
23,385.017),(2024,304.334),(2025,299.6) )
)
Cause : Time
Cause : one year
one year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Makes Time dimensionless in lookup functions
Equations : one year = 1
Use : Deep water oxygen
Use : Observed catch
Use : Observed survey index
Use : Observed zooplankton
Use : Phosphorus concentration
Oxygen effect
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Whitefish eggs lie on the lake bottom over winter and die if oxygen is low (IGKB, 2014). 1 at the reference oxygen level.
Equations : Oxygen effect =
( Deep water oxygen
/ ( Deep water oxygen
+ Oxygen half saturation ) )
/ ( Reference oxygen
/ ( Reference oxygen
+ Oxygen half saturation ) )

Cause : Deep water oxygen
Cause : Oxygen half saturation
Cause : Reference oxygen
Use : Recruitment
Oxygen half saturation
Units : mgO2/L
Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION: oxygen level at which egg survival is halved.
Equations : Oxygen half saturation = 2
Use : Oxygen effect
Phosphorus concentration
Units : ugP/L
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : DATA: total phosphorus, annual mean (IGKB/BOWIS) 1960-2025. 1950 = 7 ugP/L is an ASSUMPTION (before eutrophication). Held at the 2025 value afterwards.
Equations : Phosphorus concentration =
WITH LOOKUP( Time
/ one year ,
((1950,7),(1960,15),(1961,19),(1962,24),(1963,30),(1964,34),(1965,36),(1966,35),(1967,45.1),(1968,43.7),(1969,47),(1970,52.5),(1971,52.6),(1972,69.6),(1973,76.5),(1974,78.9),(1975,80.1),(1976,83.6),(1977,83.9),(1978,77.8),(1979,81.4),(1980,78.7),(1981,78.5),(1982,73.8),(1983,70.9),(1984,64.7),(1985,62.1),(1986,58.7),(1987,52.4),(1988,42.1),(1989,40.1),(1990,36.3),(1991,32.5),(1992,29.9),(1993,26.2),(1994,24.8),(1995,22.7),(1996,19),(1997,16.7),(1998,15.4),(1999,14),(2000,12.8),(2001,11.6),(2002,10.7),(2003,10),(2004,8.9),(2005,8.5),(2006,7.6),(2007,7.5),(2008,7.6),(2009,6.4),(2010,5.8),(2011,5.9),(2012,6.3),(2013,6.7),(2014,6.9),(2015,7.6),(2016,8),(2017,7.5),(2018,7.6),(2019,6.2),(2020,6.3),(2021,6),(2022,6.2),(2023,5.4),(2024,6.1),(2025,5.3) )
)
Cause : Time
Cause : one year
Use : Zooplankton growth
Recruitment
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : New young whitefish per year: more adults -> more offspring (R1). Depends on food (zooplankton), deep-water oxygen and sticklebacks eating eggs and larvae.
Equations : Recruitment =
Adult whitefish
* Max recruitment rate
* Food effect
* Oxygen effect
* Stickleback egg effect
Cause : Adult whitefish
Cause : Food effect
Cause : Max recruitment rate
Cause : Oxygen effect
Cause : Stickleback egg effect
Use : Young whitefish
Reference oxygen
Units : mgO2/L
Variable type : Constant
Group : .whitefish model (1)
Comment : Oxygen above 6 mg/L is considered a good supply of the deep water (IGKB, 2018).

Equations : Reference oxygen = 6
Use : Oxygen effect
Reference zooplankton
Units : kInd/m2
Variable type : Constant
Group : .whitefish model (1)
Comment : Reference level, about today's zooplankton density (IGKB).
Equations : Reference zooplankton = 300
Use : Food effect
SAVEPER
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Output interval.
Equations : SAVEPER = 1
Scenario start year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : From this year on the scenario values apply.
Equations : Scenario start year = 2027
Use : Stickleback pressure
Stickleback decline
Units : Dmnl
Variable type : Constant
Group : .whitefish model (1)
Comment : Decline of about 80 % (IGKB, n.d.-a). SCENARIO 0 = no decline.
Equations : Stickleback decline = 0.8
Use : Stickleback pressure
Stickleback decline year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : The stickleback population declined in 2024 (IGKB, n.d.-a).
Equations : Stickleback decline year = 2024
Use : Stickleback pressure
Stickleback egg effect
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Share of whitefish eggs and larvae that are not eaten by sticklebacks.
Equations : Stickleback egg effect =
1
- Stickleback egg impact
* Stickleback pressure
Cause : Stickleback egg impact
Cause : Stickleback pressure
Use : Recruitment
Stickleback egg impact
Units : Dmnl
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: share of whitefish eggs/larvae eaten at full stickleback pressure.
Equations : Stickleback egg impact = 0.505776
Use : Stickleback egg effect
Stickleback grazing rate

Units : 1/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: share of zooplankton eaten by sticklebacks per year at full stickleback pressure.
Equations : Stickleback grazing rate = 0.203165
Use : Grazing by sticklebacks
Stickleback pressure
Units : Dmnl
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Stickleback index: 0 before 2012, 1 from 2012 (spread into open water), reduced by 80 % from 2024, scenario value from 2027.
Equations : Stickleback pressure =
IF THEN ELSE ( Time
< Stickleback start year ,
0,
IF THEN ELSE ( Time
< Stickleback decline year ,
1,
IF THEN ELSE ( Time
< Scenario start year ,
1
- Stickleback decline ,
Stickleback pressure from 2027 ) ) )
Cause : Time
Cause : Scenario start year
Cause : Stickleback decline
Cause : Stickleback decline year
Cause : Stickleback pressure from 2027
Cause : Stickleback start year
Use : Grazing by sticklebacks
Use : Stickleback egg effect
Stickleback pressure from 2027
Units : Dmnl
Variable type : Constant
Group : .whitefish model (1)
Comment : SCENARIO: 0.2 = stays low, 1 = renewed stickleback pressure.
Equations : Stickleback pressure from 2027 = 0.2
Use : Stickleback pressure
Stickleback start year
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Sticklebacks spread into the open water in 2012 (Wasser-Agenda 21, 2023; SRF, 2024).
Equations : Stickleback start year = 2012
Use : Stickleback pressure
Survey catchability
Units : Dmnl/t
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: scales the model stock to the survey index.
Equations : Survey catchability = 0.0156565
Use : Model survey index
Time

Units : Year
Variable type : Time Base
Group : error
Range : Min:0, Max:1
Comment : Internally defined simulation time.
Equations : Time =
INTEG( 1,
INITIAL TIME )
Cause : INITIAL TIME
Use : Catch
Use : Deep water oxygen
Use : Observed catch
Use : Observed survey index
Use : Observed zooplankton
Use : Phosphorus concentration
Use : Stickleback pressure
TIME STEP
Units : Year
Variable type : Constant
Group : .whitefish model (1)
Comment : Euler integration step.
Equations : TIME STEP = 0.0625
Whitefish grazing rate
Units : 1/(Year*t)
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: share of zooplankton eaten per t of adult whitefish and year.
Equations : Whitefish grazing rate = 0.000626102
Use : Grazing by whitefish
Young mortality
Units : t/Year
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Young fish that die before they mature.
Equations : Young mortality =
Young whitefish
* Young mortality rate
Cause : Young whitefish
Cause : Young mortality rate
Use : Young whitefish
Young mortality rate
Units : 1/Year
Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION.
Equations : Young mortality rate = 0.3
Use : Young mortality
Young whitefish
Units : t
Variable type : Stock
Group : .whitefish model (1)
Comment : STOCK: young whitefish (eggs, larvae, juveniles) that are not yet catchable, in tonnes of future adult biomass.
Equations : Young whitefish =
INTEG( Recruitment
- Maturation

- Young mortality ,
Initial young )
Cause : Initial young
Cause : Maturation
Cause : Recruitment
Cause : Young mortality
Use : Maturation
Use : Young mortality
Zooplankton
Units : kInd/m2
Variable type : Stock
Group : .whitefish model (1)
Comment : STOCK: crustacean zooplankton (whitefish food), in 1000 individuals per m2.
Equations : Zooplankton =
INTEG( Zooplankton growth
- Zooplankton mortality
- Grazing by whitefish
- Grazing by sticklebacks ,
Initial zooplankton )
Cause : Grazing by sticklebacks
Cause : Grazing by whitefish
Cause : Initial zooplankton
Cause : Zooplankton growth
Cause : Zooplankton mortality
Use : Food effect
Use : Grazing by sticklebacks
Use : Grazing by whitefish
Use : Zooplankton mortality
Zooplankton growth
Units : kInd/(Year*m2)
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : More phosphorus -> more algae -> more zooplankton (algae are not modelled separately).
Equations : Zooplankton growth =
Max zooplankton production
* Phosphorus concentration
/ ( Phosphorus concentration
+ Zooplankton P half saturation )
Cause : Max zooplankton production
Cause : Phosphorus concentration
Cause : Zooplankton P half saturation
Use : Zooplankton
Zooplankton mortality
Units : kInd/(Year*m2)
Variable type : Auxiliary
Group : .whitefish model (1)
Comment : Natural death of zooplankton.
Equations : Zooplankton mortality =
Zooplankton
* Zooplankton mortality rate
Cause : Zooplankton
Cause : Zooplankton mortality rate
Use : Zooplankton
Zooplankton mortality rate
Units : 1/Year

Variable type : Constant
Group : .whitefish model (1)
Comment : ASSUMPTION: zooplankton turns over quickly (several generations per year).
Equations : Zooplankton mortality rate = 2
Use : Zooplankton mortality
Zooplankton P half saturation
Units : ugP/L
Variable type : Constant
Group : .whitefish model (1)
Comment : CALIBRATED: phosphorus level at which zooplankton production is half its maximum.
Equations : Zooplankton P half saturation = 21.309
Use : Zooplankton growth